BESHOY HABIB

Entry-Level Controls, Automation & Embedded Systems Engineer

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EDUCATION

Bachelor of Science, Electromechanical Systems Engineering Technology **May 2026**

California State Polytechnic University, Pomona, CA | Dean's List – Spring 2025

Relevant Coursework: Embedded Systems Design, Microcontrollers, Control Systems, Circuit Analysis, Robotics and Automation, Mechatronics, Power Electronics, Electrical Machines.

EXPERIENCE

CNC Engineering Intern | *Al Saqr Steel* **Aug 2023 – Jan 2024**

- Programmed, set up, and operated CNC machines for automated precision manufacturing applications.
- Diagnosed and resolved machine setup and tooling issues, improving part tolerance consistency and reducing defect rates on production runs.
- Improved production timing and increased manufacturing productivity by optimizing machine setup procedures, **reducing equipment downtime by approximately 40–50%**.
- Performed routine maintenance and quality verification checks, helping maintain consistent output quality across manufacturing shifts.

Electronics Supply Technician | *HD Electronics* **Jan 2023 – Jun 2023**

- **Managed inventory of 100–300 electronic components**, tracking stock levels and reordering to prevent part shortages.
- Supported several engineers with part selection and sourcing, streamlining procurement for ongoing projects.
- Organized component inventory system, improving retrieval speed and reducing time spent locating parts.

ACADEMIC & TECHNICAL PROJECTS

Autonomous Robotic Waste-Sorting System — ASME Student Design Competition (AY 2025–2026)

- **Represented Cal Poly Pomona at the national ASME Student Design Competition** in Florida, designing and building an autonomous robotic waste-sorting system from concept through live demonstration.
- Designed the mechanical structure and CAD layout, integrating microcontrollers, RGB color sensors, and wireless communication for automated waste classification.
- **Cut sensor calibration and overall system testing time by roughly half** by streamlining the testing process, and improved the sorting mechanism's design for more reliable object handling.
- Integrated mechanical, electrical, and software systems into a single working autonomous prototype, from CAD design through embedded control code.
- Developed and tested C/C++ control algorithms for sensor interfacing, autonomous navigation, and object detection, demonstrating the full system live at competition.

Computer Numerical Control (CNC) Manufacturing (Spring 2024)

- **Led a team of 5 student engineers** in building a CNC machine from scratch, then programmed, set up, and operated it for automated precision manufacturing applications.
- Performed maintenance, troubleshooting, and setup verification, refining processes to improve part quality and reduce setup time.

Custom 3D Printer (Fall 2023)

- **Led a team of 5 student engineers** in designing, assembling, and calibrating a 3D printer built entirely from scratch, used to manufacture precision parts for CNC machine assemblies.
- Tuned printing parameters and closed-loop calibration settings, reducing print defects and improving dimensional accuracy.

Robotic Vehicle with Automated Gripping Arm (Spring 2022)

- **Designed and built an Arduino-based robotic vehicle with an automated gripping arm** for object handling and transport.
- Integrated sensors, actuators, and embedded control systems, programming microcontroller-based control algorithms in C/C++.
- Iterated on the gripping mechanism design to improve grip reliability and object-handling accuracy.

TECHNICAL SKILLS

Programming: C/C++, Embedded C, Python, MATLAB, G-code Programming

Embedded & Controls: Arduino, Microcontrollers, PLC Programming & Fundamentals, Sensor & Actuator Integration, Control Systems, System Modeling

Design & Manufacturing: SolidWorks, AutoCAD, CNC Machining, Quality Control, Hardware-Software Integration